

The Relativistic–Thermodynamic General Equilibrium

Planetary Geometry, Geological Depth, Astronomical Distance,
Causal Markets, and the Economics of Physical Frontiers

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Abstract

A civilization located on the surface of a spherical planet occupies a remarkable economic position. At every point on the planetary surface there exist two opposite radial directions. Geological investigation proceeds approximately along the inward normal toward the planetary interior, whereas astronomical investigation proceeds approximately along the outward normal toward extraterrestrial space. The resulting geometry suggests a general equilibrium theory in which the planetary surface is a scarce interface between two physically constrained opportunity domains.

The inward geological domain is not economically accessible to the geometric center. Increasing temperature, pressure, material failure, drilling costs, energy requirements, and other geophysical constraints generate a thermodynamic–mechanical accessibility frontier strictly shallower than the planetary radius. The outward astronomical domain is likewise not economically equivalent to an unbounded Euclidean opportunity set. Special relativity, finite signal speed, propulsion costs, radiation, biological survival, discounting, information delay, and cosmological causal structure generate technological, economic, and causal horizons.

This paper develops a Relativistic–Thermodynamic General Equilibrium (RTGE) framework in which civilization occupies a dynamically evolving economic shell around a spherical planetary surface. We derive resource constraints, shadow prices, Hotelling-type geological scarcity conditions, delayed astronomical arbitrage conditions, welfare criteria, endogenous research allocation, political and strategic distortions, and conditions for radial structural transformation. We show that sufficiently spatially extended economies cannot in general possess an instantaneous Walrasian equilibrium defined with respect to a universal operational information set. Instead, equilibrium becomes spacetime-indexed and information is necessarily retarded by finite signal propagation.

The framework integrates economics, finance, geology, geophysics, astronomy, relativity, thermodynamics, chemistry, biology, psychology, political science, international relations, warfare economics, statistics, computer science, artificial intelligence, and philosophy into a common constrained optimization architecture.

The paper ends with “The End”

Keywords: general equilibrium; relativity; thermodynamics; geology; astronomy; planetary economics; resource scarcity; causal horizon; Hotelling rule; space economics; delayed arbitrage; welfare economics; geopolitics; artificial intelligence.

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1 Introduction

Consider a spherical planet of radius R . Let its surface be

$$S^2(R) = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| = R\}.$$

At every surface point $\mathbf{x}$, define the outward unit normal

$$\mathbf{n}(\mathbf{x}) = \frac{\mathbf{x}}{R}.$$

There are then two distinguished radial directions:

$$+\mathbf{n}(\mathbf{x})$$

and

$$-\mathbf{n}(\mathbf{x}).$$

Astronomical observation and extraterrestrial expansion proceed approximately in the $+\mathbf{n}$ direction. Geological investigation and subsurface extraction proceed approximately in the $-\mathbf{n}$ direction.

This elementary geometric fact produces a useful economic decomposition:

$$\boxed{\text{Planetary interior} \longleftarrow S^2(R) \longrightarrow \text{Extraterrestrial exterior.}}$$

The surface is neither merely land nor merely a boundary. It is the economic interface through which civilization accesses both domains.

The central thesis of this paper is that neither radial opportunity set should be represented by unconstrained Euclidean geometry.

The geological frontier is restricted by thermodynamics, pressure, material strength, extraction technology, chemistry, and economic cost.

The astronomical frontier is restricted by energy, propulsion, finite signal speed, relativity, biological survival, discounting, and cosmological causal structure.

Consequently, the economically relevant radial domain at date t is

$$\boxed{-D_G^*(t) \leq z \leq D_A^*(t),}$$

where $z = 0$ denotes the planetary surface, $D_G^*(t)$ is the economically accessible geological depth, and $D_A^*(t)$ is the economically accessible astronomical distance.

The resulting system will be called the *Relativistic-Thermodynamic General Equilibrium*.

2 Planetary Geometry

The planetary volume is

$$V_P = \frac{4}{3}\pi R^3,$$

and surface area is

$$S_P = 4\pi R^2.$$

Therefore,

$$\frac{V_P}{S_P} = \frac{R}{3}.$$

This ratio has an immediate economic interpretation: larger planets possess more interior volume per unit of surface interface.

Let $d_G \geq 0$ denote depth below the surface. Then radial distance from the center is

$$r = R - d_G.$$

Let $d_A \geq 0$ denote outward distance from the surface. The corresponding radial coordinate is

$$r = R + d_A.$$

Equivalently, define signed radial distance z by

$$z = \begin{cases} -d_G, & \text{inside the planet,} \\ 0, & \text{on the surface,} \\ d_A, & \text{outside the planet.} \end{cases}$$

The purely geometric radial domain would be

$$-R \leq z < \infty.$$

The economically accessible domain will be much smaller.

3 The Inward and Outward Normals

Definition 1 (Inward opportunity direction). *At $\mathbf{x} \in S^2(R)$, the inward opportunity direction is*

$$-\mathbf{n}(\mathbf{x}).$$

It represents geological, mining, geothermal, subsurface engineering, geochemical, and related activities.

Definition 2 (Outward opportunity direction). *At $\mathbf{x} \in S^2(R)$, the outward opportunity direction is*

$$+\mathbf{n}(\mathbf{x}).$$

It represents astronomical observation, orbital activity, communication, remote sensing, space transportation, and extraterrestrial production.

Locally the directions are geometrically symmetric:

$$\|+\mathbf{n}\| = \|-\mathbf{n}\| = 1.$$

But the physical constraints encountered along them are radically different.

4 A Geometric Representation

Figure 1 illustrates the basic architecture.

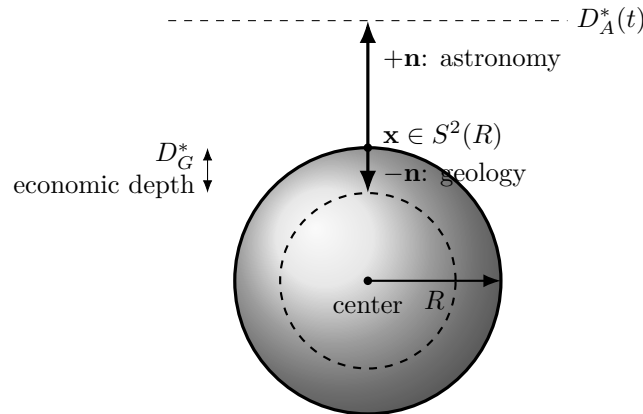


Figure 1: The planetary surface as an interface between inward geological and outward astronomical opportunity domains. The economically accessible geological frontier lies strictly above the center, while the astronomical frontier is dynamically determined by technology, economics, and causal constraints.

5 Geological Thermodynamics

Let subsurface temperature at depth d be $T(d)$.

For the basic theory assume

$$T'(d) > 0$$

over the economically relevant depth interval.

A local approximation is

$$T(d) = T_0 + g_T d,$$

where T_0 is surface temperature and $g_T > 0$ is a geothermal gradient.

This linear specification is only a local approximation; actual planetary temperature profiles depend on conduction, convection, radioactive heating, composition, phase transitions, mantle dynamics, and planetary history.

Let $T_{\max}$ denote the maximum temperature compatible with a particular technology.

Define D_T implicitly by

$$T(D_T) = T_{\max}.$$

For the linear approximation,

$$D_T = \frac{T_{\max} - T_0}{g_T}.$$

Physical feasibility requires

$$D_T < R.$$

Thus temperature alone may prevent economically relevant penetration anywhere near the center.

6 Pressure, Chemistry, and Materials

Temperature is not the only geological constraint.

Let pressure satisfy

$$P = P(d), \quad P'(d) > 0.$$

Let $P_{\max}$ denote a technological pressure limit. Define

$$P(D_P) = P_{\max}.$$

Chemical environments may also become corrosive or reactive. Let

$$\chi(d)$$

represent an index of chemical hostility, including corrosion, reactive fluids, phase instability, or contamination risk.

Let

$$\chi(D_\chi) = \chi_{\max}.$$

Likewise, define D_M as the depth at which material-strength constraints become binding.

The technologically feasible geological frontier is therefore

$$D_G^{\text{tech}} = \min\{D_T, D_P, D_\chi, D_M, \dots\}.$$

Economic feasibility may impose a still tighter restriction.

7 The Economic Geological Frontier

Let p_G denote the output price of a geological resource and let $q_G(d)$ denote recoverable output at depth d .

Let extraction cost be

$$C_G = C_G(q_G, d, T(d), P(d), \chi(d)).$$

Assume

$$\frac{\partial C_G}{\partial d} > 0,$$

$$\frac{\partial C_G}{\partial T} > 0,$$

and, over the relevant region,

$$\frac{\partial C_G}{\partial P} > 0.$$

Then total differentiation gives

$$\frac{dC_G}{dd} = \frac{\partial C_G}{\partial d} + \frac{\partial C_G}{\partial T} T'(d) + \frac{\partial C_G}{\partial P} P'(d) + \frac{\partial C_G}{\partial \chi} \chi'(d).$$

Under positive marginal effects,

$$\frac{dC_G}{dd} > 0.$$

The economically optimal geological depth D_G^* therefore satisfies

$$D_G^* \leq D_G^{\text{tech}} < R.$$

Proposition 1 (Finite geological accessibility). *Suppose $R < \infty$, $T'(d) > 0$, extraction technology has finite operating limits, and extraction costs become prohibitive no later than some depth $D < R$. Then the economically accessible geological domain is bounded by*

$$0 \leq d_G \leq D_G^* < R.$$

Proof. The physical planetary domain requires $d_G \leq R$. By assumption at least one technological or economic constraint becomes binding at $D < R$. Therefore feasible extraction cannot extend beyond D . Since the economically optimal set is a subset of the technologically feasible set,

$$D_G^* \leq D < R.$$

Hence $D_G^* < R$. □

8 Accessible Geological Volume

The accessible geological shell occupies

$$R - D_G^* \leq r \leq R.$$

Its volume is

$$V_G^* = \frac{4\pi}{3} [R^3 - (R - D_G^*)^3].$$

Expanding,

$$V_G^* = 4\pi R^2 D_G^* - 4\pi R (D_G^*)^2 + \frac{4\pi}{3} (D_G^*)^3.$$

For

$$D_G^* \ll R,$$

the first-order approximation is

$$V_G^* \approx 4\pi R^2 D_G^*.$$

Hence accessible geological volume is approximately surface area multiplied by accessible depth.

9 Finite Geological Stocks

Let $\rho_G(r)$ denote economically relevant resource density.

The total physically existing stock is

$$Q_G^{\text{physical}} = 4\pi \int_0^R \rho_G(r) r^2 dr.$$

The accessible stock is instead

$$Q_G^{\text{acc}} = 4\pi \int_{R-D_G^*}^R \rho_G(r) r^2 dr.$$

Generally,

$$Q_G^{\text{acc}} < Q_G^{\text{physical}}.$$

This distinction separates geological existence from economic accessibility.

10 Hotelling Scarcity

Let $X_G(t)$ denote the remaining accessible exhaustible stock.

Then

$$\dot{X}_G(t) = -q_G(t).$$

In the simplest deterministic competitive resource model, the scarcity rent $\mu_G(t)$ satisfies

$$\dot{\mu}_G = r_f \mu_G,$$

where r_f is the relevant risk-free return.

Thus

$$\frac{\dot{\mu}_G}{\mu_G} = r_f.$$

With extraction costs MC_G , the resource price may be decomposed as

$$p_G = MC_G + \mu_G.$$

The geological price therefore incorporates both contemporaneous extraction cost and intertemporal scarcity.

11 Astronomical Accessibility

Now consider the outward direction.

Pure Euclidean geometry would permit

$$0 \leq d_A < \infty.$$

But this is not the economically relevant domain.

Let

$$D_A^{\text{tech}}(t)$$

denote the maximum technologically reachable scale at date t .
Constraints include:

- propulsion energy;
- achievable velocity;
- spacecraft reliability;
- radiation;
- communication;
- life support;
- biological lifetime;
- capital depreciation;
- navigation;
- computation;
- political durability;
- economic discounting.

Therefore,

$$D_A^*(t) \leq D_A^{\text{tech}}(t).$$

12 Relativistic Constraints

Special relativity imposes the causal speed limit

$$v < c.$$

Define

$$\beta = \frac{v}{c}.$$

Then

$$0 \leq \beta < 1.$$

The Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}.$$

As $v \rightarrow c$,

$$\gamma \rightarrow \infty.$$

For a massive object, relativistic kinetic energy is

$$K = (\gamma - 1)mc^2.$$

Consequently,

$$\lim_{v \rightarrow c} K = \infty.$$

Thus a massive spacecraft cannot be accelerated to the speed of light with finite energy.

13 Communication Delay

For a stationary flat-spacetime approximation over distance D , the one-way signal delay obeys

$$\tau_s(D) \geq \frac{D}{c}.$$

A simple round-trip lower bound is

$$\tau_R(D) \geq \frac{2D}{c}.$$

Consequently, economic information cannot propagate instantaneously.

If $p_j(t)$ is a price at distant location j , location i cannot generally observe $p_j(t)$ contemporaneously. Instead its information set contains a retarded observation approximately of the form

$$p_j(t - \tau_{ij}).$$

This destroys the operational meaning of instantaneous global price equalization across sufficiently separated markets.

14 Relativistic Discounting and Economic Distance

Suppose a project at distance D generates payoff $X(D)$ after economically relevant delay $\tau(D)$.

Its discounted value is

$$PV(D) = e^{-\rho\tau(D)}X(D),$$

where $\rho > 0$ is the social or private discount rate.

Under the simple round-trip bound,

$$\tau(D) \geq \frac{2D}{c},$$

and therefore

$$PV(D) \leq X(D)e^{-2\rho D/c}.$$

If $X(D)$ grows more slowly than $e^{2\rho D/c}$, then

$$\lim_{D \rightarrow \infty} PV(D) = 0.$$

Proposition 2 (Discount-induced astronomical horizon). *Suppose*

$$PV(D) = X(D)e^{-\rho\tau(D)},$$

$$\rho > 0,$$

$$\tau(D) \geq \frac{2D}{c},$$

and

$$\lim_{D \rightarrow \infty} X(D)e^{-2\rho D/c} = 0.$$

Then arbitrarily distant opportunities have vanishing present value.

Proof. Since

$$\tau(D) \geq \frac{2D}{c},$$

we have

$$e^{-\rho\tau(D)} \leq e^{-2\rho D/c}.$$

Hence

$$0 \leq PV(D) \leq X(D)e^{-2\rho D/c}.$$

The assumed limiting condition and the squeeze theorem imply

$$\lim_{D \rightarrow \infty} PV(D) = 0.$$

□

This does not imply that physical space itself possesses a finite edge. Rather, it implies that the economically actionable opportunity set may be effectively bounded.

15 Causal, Observable, Reachable, and Economic Domains

Cosmology requires several domains to be distinguished carefully.

Let

$$\mathcal{A}_{\text{obs}}$$

denote the observable domain,

$$\mathcal{A}_{\text{causal}}$$

the relevant causal domain,

$$\mathcal{A}_{\text{reach}}$$

the domain physically reachable under specified propulsion and cosmological conditions, and

$$\mathcal{A}_{\text{econ}}$$

the economically exploitable domain.

Under suitable definitions one expects

$$\boxed{\mathcal{A}_{\text{econ}} \subseteq \mathcal{A}_{\text{reach}} \subseteq \mathcal{A}_{\text{causal}}}$$

with observational accessibility requiring separate treatment because “observable,” “reachable,” and “controllable” are not identical concepts in relativistic cosmology.

The economically relevant astronomical frontier can therefore be represented abstractly as

$$D_A^* = \min\{D_{\text{prop}}, D_{\text{energy}}, D_{\text{bio}}, D_{\text{comm}}, D_{\text{causal}}, D_{\text{econ}}\}.$$

16 The Planetary Economic Shell

Combining both sides gives

$$\boxed{-D_G^*(t) \leq z \leq D_A^*(t)}.$$

The surface $z = 0$ separates the two physical regimes.

Definition 3 (Planetary economic shell). *The planetary economic shell at time t is*

$$\mathcal{S}_E(t) = \{(z, \theta, \phi) : -D_G^*(t) \leq z \leq D_A^*(t)\},$$

subject to physical, technological, causal, institutional, and economic feasibility.

The shell is endogenous because technology can change both boundaries.

17 Two-Sector Production

Let geological output be

$$Y_G = F_G(K_G, L_G, S_G, X_G, A_G, A_A),$$

and astronomical output be

$$Y_A = F_A(K_A, L_A, S_A, X_A, A_A, A_G).$$

Here:

- K_i denotes capital;
- L_i denotes labor;
- S_i denotes surface access;
- X_i denotes sector-specific resources;
- A_i denotes sector-specific knowledge.

Capital feasibility requires

$$K_G + K_A + K_O = \bar{K},$$

labor feasibility requires

$$L_G + L_A + L_O = \bar{L},$$

and surface feasibility requires

$$S_G + S_A + S_O \leq 4\pi R^2.$$

The residual sector O contains agriculture, housing, manufacturing, ecosystem preservation, government, and other uses of the surface.

18 The Shadow Price of the Surface

Let λ_S denote the Lagrange multiplier on

$$S_G + S_A + S_O \leq 4\pi R^2.$$

Then

$$\lambda_S = \frac{\partial W^*}{\partial (4\pi R^2)}$$

is the marginal welfare value of additional usable planetary surface.

Surface land therefore possesses an option value arising from access to both radial domains.

A site may support:

- mines;
- boreholes;
- geothermal installations;
- geological observatories;
- telescopes;
- radio arrays;
- launch facilities;

- radar;
- satellite control;
- spaceports.

Land rents can therefore capitalize radial access.

19 Cross-Sector Complementarity

Geology and astronomy are substitutes for scarce labor and capital but complements in knowledge and materials.

Suppose

$$\frac{\partial Y_A}{\partial A_G} > 0$$

and

$$\frac{\partial Y_G}{\partial A_A} > 0.$$

Geological knowledge assists astronomy through materials science, mineral supply, geodesy, geochemistry, and planetary science.

Astronomical technology assists geology through remote sensing, satellite navigation, gravimetry, spectroscopy, imaging, and communications.

Thus the two sectors generate reciprocal knowledge spillovers.

20 Competitive Equilibrium

If capital and labor are mobile and markets are locally competitive, an interior equilibrium approximately requires

$$MPK_G - \delta_G = MPK_A - \delta_A = r,$$

and

$$MPL_G = MPL_A = w,$$

after correcting for sector-specific risk, taxes, subsidies, and adjustment costs.

For surface access,

$$p_G \frac{\partial F_G}{\partial S_G} = p_A \frac{\partial F_A}{\partial S_A} = \lambda_S$$

at an undistorted interior allocation.

21 The Social Planner

Consider

$$\max \int_0^\infty e^{-\rho t} U(C_t, H_t, E_t, K_t) dt,$$

where H_t denotes scientific knowledge and E_t environmental quality.

The planner faces

$$C_t + I_{G,t} + I_{A,t} + I_{O,t} + I_{R,t} = Y_G(t) + Y_A(t) + Y_O(t),$$

$$\dot{K}_i = I_i - \delta_i K_i,$$

$$\dot{X}_G = -q_G,$$

and

$$S_G + S_A + S_O \leq 4\pi R^2.$$

Research investment I_R can move the frontiers:

$$\dot{D}_G^* = \Phi_G(I_G^R, A_G, D_G^*),$$

$$\dot{D}_A^* = \Phi_A(I_A^R, A_A, D_A^*).$$

The planner compares the marginal social values

$$\frac{\partial W}{\partial D_G^*} \frac{\partial D_G^*}{\partial I_G^R}$$

and

$$\frac{\partial W}{\partial D_A^*} \frac{\partial D_A^*}{\partial I_A^R}.$$

At an interior research allocation,

$$\boxed{\frac{\partial W}{\partial D_G^*} \frac{\partial D_G^*}{\partial I_G^R} = \frac{\partial W}{\partial D_A^*} \frac{\partial D_A^*}{\partial I_A^R}.$$

22 The Inward–Outward Asymmetry Theorem

Theorem 1 (Relativistic–thermodynamic radial asymmetry). *Suppose:*

1. *civilization occupies $S^2(R)$;*
2. *geological activity proceeds inward;*
3. *geological accessibility is restricted by increasing thermodynamic, mechanical, and economic costs;*
4. *astronomical activity proceeds outward;*
5. *outward communication and transportation obey relativistic causality;*
6. *economic activity requires finite technological resources and positive net present value.*

Then at every finite date the economically relevant radial opportunity set can be represented by finite endogenous frontiers

$$-D_G^*(t) \leq z \leq D_A^*(t),$$

where

$$D_G^*(t) < R$$

under a binding sub-central geological constraint and

$$D_A^*(t) < \infty$$

whenever technological, causal, biological, or economic constraints impose a finite effective outward frontier.

Proof. The geological result follows from the existence of a binding constraint $D < R$, giving $D_G^* \leq D < R$.

For the outward domain, finite-date technology implies finite transport and communication capability. Relativistic causality prevents superluminal information or massive-object transportation. Positive discounting and finite resource budgets further restrict economically valuable operations. If any such restriction produces a finite binding distance, then

$$D_A^*(t) < \infty.$$

Combining the two inequalities yields the bounded economic shell. $\square$

Remark 1. *The theorem does not assert that spacetime itself must have finite spatial extent. It concerns the economically actionable opportunity set of a civilization at a given date.*

23 Relativistic General Equilibrium

Ordinary Walrasian equilibrium often suppresses communication delay.

Suppose markets i and j are separated by distance D_{ij} .

Then

$$\tau_{ij} \geq \frac{D_{ij}}{c}.$$

Hence the information set at i cannot contain arbitrarily recent information from j .

Let

$$\mathcal{I}_i(t)$$

denote information available at market i .

Then distant information satisfies approximately

$$p_j(t - \tau_{ij}) \in \mathcal{I}_i(t),$$

rather than necessarily

$$p_j(t) \in \mathcal{I}_i(t).$$

Therefore the standard instantaneous law of one price,

$$p_i(t) = p_j(t),$$

must be replaced by a dynamically consistent arbitrage relation.

One generic formulation is

$$p_i(t) = e^{-r\tau_{ij}} \mathbb{E}_t [p_j(t + \tau_{ij})] - C_{ij} - \Pi_{ij},$$

where C_{ij} is transport or communication cost and Π_{ij} collects risk, liquidity, default, political, and technological premia.

24 A No-Instantaneous-Global-Information Result

Proposition 3 (No instantaneous operational global information). *Suppose two economic agents are spatially separated by $D > 0$ and all information transmission is causal. Then neither agent can condition current actions on newly generated contemporaneous information at the other location.*

Proof. Causality requires a signal transmission time satisfying

$$\tau \geq \frac{D}{c} > 0.$$

Therefore information generated at one location at time t cannot arrive at the other at the same local time t . Hence the receiving agent's operational information set excludes that newly generated contemporaneous information. $\square$

This proposition does not rule out mathematical equilibrium concepts defined globally. It rules out interpreting such equilibria as literally instantaneously coordinated by physically transmitted information.

25 Finance Across Astronomical Distances

Relativistic delay modifies finance.

Consider an asset paying $X_{t+\tau}$ at a distant location.

A stylized price is

$$P_t = \mathbb{E}_t [M_{t,t+\tau} X_{t+\tau}],$$

where $M_{t,t+\tau}$ is the stochastic discount factor.

As τ increases:

- discounting becomes more important;
- model uncertainty accumulates;
- counterparty risk rises;
- political continuity becomes less certain;
- technological obsolescence becomes more likely;
- information becomes increasingly asynchronous.

A reduced-form risk-adjusted discount factor may be written

$$M_{t,t+\tau} = \exp \left[- \int_t^{t+\tau} (r_s + \pi_s^R + \pi_s^P + \pi_s^T) ds \right],$$

where π^R , π^P , and π^T denote relativistic-distance, political, and technological risk premia.

26 Chemistry and Materials

The inward frontier depends strongly on chemical and materials science.

At greater depth, relevant processes include:

- high-temperature reactions;
- phase transitions;
- corrosion;
- pressure-dependent mineral phases;
- diffusion;
- fluid-rock interaction;
- changes in viscosity;
- creep and plastic deformation.

Let material survival probability be

$$P_M(d, t) = \exp[-h_M(d)t],$$

where $h_M(d)$ is a depth-dependent hazard rate satisfying

$$h'_M(d) > 0$$

over the relevant range.

Then expected extraction value becomes

$$\mathbb{E}[\Pi_G(d)] = P_M(d, t)\Pi_G(d).$$

Thus materials degradation can create an economic frontier even before an absolute physical impossibility is reached.

27 Biological Constraints

Human biological systems impose additional constraints on astronomical expansion.

Let cumulative mission hazard be

$$H_A(D) = H_{\text{rad}}(D) + H_{\text{micro}}(D) + H_{\text{life}}(D) + H_{\text{psych}}(D).$$

A reduced-form survival probability is

$$P_S(D) = e^{-H_A(D)}.$$

Expected mission value is then

$$EV_A(D) = P_S(D)X_A(D) - C_A(D).$$

The economic frontier may satisfy

$$EV_A(D_A^*) = 0.$$

Robotics and autonomous AI can alter this constraint by reducing direct dependence on biological survival.

28 Psychology and Psychiatry of Extreme Distance

Long-duration isolation, confinement, communication delay, disrupted social networks, monotony, risk exposure, and institutional separation may affect human performance.

Let effective labor be

$$L_A^{\text{eff}} = L_A \Psi(D, \tau, \sigma, I),$$

where

- τ is communication delay;
- σ is environmental stress;
- I measures isolation;
- Ψ is a performance function.

A plausible reduced form has

$$\frac{\partial \Psi}{\partial \tau} < 0, \quad \frac{\partial \Psi}{\partial \sigma} < 0, \quad \frac{\partial \Psi}{\partial I} < 0.$$

These are population-level modeling assumptions, not claims about inevitable individual psychiatric outcomes.

Autonomy, social design, habitat architecture, medical support, and AI systems may increase Ψ .

29 Artificial Intelligence and Automation

AI changes both frontiers.

For geology, AI may improve:

- seismic inversion;
- mineral exploration;
- drilling control;
- predictive maintenance;
- robotic extraction;
- hazard forecasting.

For astronomy, AI may improve:

- autonomous navigation;
- scientific observation;
- fault recovery;
- robotic construction;
- local decision-making;
- delayed-command resilience.

Let AI capital be K_{AI} .

Then

$$D_G^* = D_G^*(K_{AI}), \quad \frac{\partial D_G^*}{\partial K_{AI}} > 0$$

over some technological range, and similarly

$$D_A^* = D_A^*(K_{AI}), \quad \frac{\partial D_A^*}{\partial K_{AI}} > 0.$$

AI therefore becomes a frontier-shifting technology.

30 The Autonomy Principle

As communication delay increases, centralized control becomes less effective.

Let central-control quality be

$$Q_C(D) = Q_0 e^{-\kappa D/c},$$

where $\kappa > 0$.

Let autonomous local-control quality be $Q_L(A_{AI})$.

Decentralization becomes optimal when

$$Q_L(A_{AI}) > Q_C(D).$$

The critical distance satisfies

$$Q_L = Q_0 e^{-\kappa D_{\text{dec}}/c}.$$

Hence

$$D_{\text{dec}} = \frac{c}{\kappa} \ln \left(\frac{Q_0}{Q_L} \right),$$

where the expression applies when $0 < Q_L < Q_0$.

Thus relativity can generate an endogenous institutional transition from centralized to locally autonomous governance.

31 Political Science and Institutional Distance

Political authority also faces finite information speed.

Suppose political control effectiveness is

$$\Gamma(D) = \Gamma_0 e^{-\alpha\tau(D)} e^{-\beta C_G(D)},$$

where $C_G(D)$ is governance cost.

Then

$$\Gamma'(D) < 0$$

under increasing delay and governance cost.

A remote settlement may become effectively autonomous when

$$\Gamma(D) < \bar{\Gamma},$$

for some minimum control threshold $\bar{\Gamma}$.

Thus astronomical distance can transform constitutional design.

Possible institutional responses include:

- delegated authority;
- federalism;
- mission command;
- local constitutional autonomy;
- precommitted contingency rules;
- machine-assisted governance.

32 Geopolitics and International Relations

On a multi-state planet, geological and astronomical frontiers create strategic competition.

Let states be indexed by $i = 1, \dots, N$.

State welfare may be

$$W_i = U_i(C_i) + \alpha_i X_i^G + \beta_i X_i^A + \gamma_i S_i - C_i^{\text{security}}.$$

Control over geological resources influences:

- industrial capacity;
- energy security;
- critical minerals;
- military production;
- strategic reserves.

Control over astronomical infrastructure influences:

- communications;
- navigation;
- intelligence;
- early warning;
- remote sensing;
- launch capability;
- orbital logistics.

The two radial domains therefore enter national power functions.

33 Warfare Economics

Let civilian output be Y_C and military output be Y_M .

A resource constraint is

$$C + I_G + I_A + M \leq Y_C + Y_G + Y_A.$$

During conflict, the shadow value of strategic geological resources may rise.

Let λ_G^P be the peacetime scarcity value and λ_G^W the wartime value.

Typically one may have

$$\lambda_G^W > \lambda_G^P$$

for militarily critical resources.

Likewise, the strategic value of satellites, communications, positioning, reconnaissance, and early-warning systems can increase sharply during war.

Thus warfare changes the relative shadow prices of both inward and outward capital.

34 A Strategic Denial Model

Suppose actor i controls infrastructure that grants rival j access to resource flow q_j .

Actor i chooses denial effort m_i .

Let

$$q_j = q_j(m_i), \quad q'_j(m_i) < 0.$$

Actor i solves

$$\max_{m_i} B_i(q_i) - \theta_i B_j(q_j(m_i)) - C_i(m_i).$$

The first-order condition is

$$-\theta_i B'_j(q_j) q'_j(m_i) = C'_i(m_i).$$

This applies to mines, ports, launch infrastructure, communications networks, orbital assets, and other strategic bottlenecks.

35 Welfare Economics

Competitive equilibrium need not be socially optimal because both sectors generate externalities.

Geological extraction can generate

$$E_G = E_{\text{pollution}} + E_{\text{ecosystem}} + E_{\text{seismic}} + E_{\text{climate}}.$$

Astronomical activity can generate

$$E_A = E_{\text{debris}} + E_{\text{spectrum}} + E_{\text{light}} + E_{\text{security}}.$$

A social welfare function can therefore be written

$$W = \int_0^\infty e^{-\rho t} [U(C_t) - \Psi_G(E_{G,t}) - \Psi_A(E_{A,t}) + \Omega(A_t)] dt.$$

Optimal Pigouvian wedges satisfy, schematically,

$$\tau_G = MD_G,$$

and

$$\tau_A = MD_A,$$

where MD_i denotes marginal external damage.

Positive knowledge spillovers may instead justify subsidies.

36 Intergenerational Welfare

The choice between extraction and exploration is inherently intergenerational.

Let generations be indexed by g .

A social objective is

$$W = \sum_{g=0}^{\infty} \beta^g U_g.$$

Aggressive geological extraction raises present consumption but lowers future resource stocks.

Astronomical research may reduce current consumption but enlarge future opportunity sets.

The allocation therefore contains a fundamental tradeoff between

consumption of inherited stocks

and

investment in expanded opportunity sets.

37 Philosophical Interpretation

The geometry supports a useful philosophical distinction.

The inward direction concerns primarily what is physically inherited:

$$-\mathbf{n} \longrightarrow \text{stock}.$$

The outward direction concerns primarily what can potentially be discovered, observed, or reached:

$$+\mathbf{n} \longrightarrow \text{opportunity}.$$

But neither stock nor opportunity is economically meaningful independently of knowledge, technology, institutions, and time.

Thus economic scarcity is not merely material scarcity. It is constrained accessibility.

A resource may physically exist yet remain outside the feasible set.

Likewise, an astronomical object may be observable but not reachable, reachable but not controllable, or technically reachable but economically worthless.

The relevant ontological hierarchy is therefore

$\text{existence} \supseteq \text{observability} \supseteq \text{technical accessibility} \supseteq \text{economic accessibility},$

with the understanding that the exact nesting depends on the object and the physical definitions employed.

38 Radial Structural Transformation

Consider the share of investment devoted to astronomy:

$$s_A(t) = \frac{I_A(t)}{I_G(t) + I_A(t)}.$$

As accessible geological stocks become scarcer,

$$\mu_G(t) \uparrow.$$

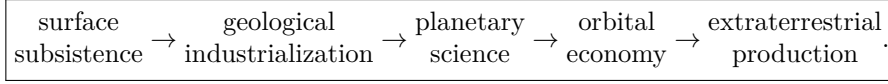
As astronomical productivity improves,

$$MPK_A(t) \uparrow.$$

A radial structural transformation occurs if

$$\dot{s}_A(t) > 0.$$

This does not imply geological activity disappears. Instead the equilibrium composition shifts. A stylized development sequence is



39 A General Equilibrium Frontier Condition

Let $V_G(D_G)$ and $V_A(D_A)$ denote welfare contributions from expanding the two frontiers. Let research costs be

$$C_R(I_G^R, I_A^R).$$

The planner chooses

$$\max_{I_G^R, I_A^R} V_G(D_G^*) + V_A(D_A^*) - C_R(I_G^R, I_A^R).$$

The first-order conditions are

$$V'_G(D_G^*) \frac{\partial D_G^*}{\partial I_G^R} = \frac{\partial C_R}{\partial I_G^R},$$

and

$$V'_A(D_A^*) \frac{\partial D_A^*}{\partial I_A^R} = \frac{\partial C_R}{\partial I_A^R}.$$

If the marginal research costs are equal,

$$\frac{\partial C_R}{\partial I_G^R} = \frac{\partial C_R}{\partial I_A^R},$$

then

$$V'_G(D_G^*) \frac{\partial D_G^*}{\partial I_G^R} = V'_A(D_A^*) \frac{\partial D_A^*}{\partial I_A^R}.$$

This is the radial research-allocation condition.

40 Comparative Statics

The model produces several immediate predictions.

If geothermal constraints become less severe through improved materials,

$$\frac{\partial D_G^*}{\partial T_{\max}} > 0.$$

If drilling productivity rises,

$$\frac{\partial D_G^*}{\partial A_G} > 0.$$

If propulsion efficiency rises,

$$\frac{\partial D_A^*}{\partial A_A} > 0.$$

If the discount rate rises,

$$\frac{\partial D_A^*}{\partial \rho} < 0$$

under conventional project-value assumptions.

If autonomous AI improves,

$$\frac{\partial D_A^*}{\partial A_{AI}} > 0.$$

If radiation risk increases,

$$\frac{\partial D_A^*}{\partial H_{\text{rad}}} < 0.$$

41 Summary of the Two Frontiers

Table 1: Inward and outward economic frontiers

Characteristic	Geological frontier	Astronomical frontier
Direction	$-\mathbf{n}$	$+\mathbf{n}$
Reference point	Planetary surface	Planetary surface
Primary physical constraint	Thermodynamics and pressure	Relativity and distance
Resource structure	Frequently exhaustible	Knowledge and extraterrestrial resources
Signal problem	Moderate locally	Increasingly severe with distance
Typical cost gradient	Depth	Distance and velocity
Biological constraint	Heat, pressure, toxicity	Radiation, isolation, duration
Economic frontier	D_G^*	D_A^*
Ultimate planetary bound	$D_G^* < R$ under assumptions	No claim of spatial edge; causal/economic bounds
Key shadow value	Depth relaxation	Distance relaxation
Strategic value	Minerals, energy, underground infrastructure	Communication, navigation, observation, mobility

42 Statistical Identification

The theory can be tested empirically.

For geological projects, define

$$Y_i^G = \alpha + \beta_1 d_i + \beta_2 T_i + \beta_3 P_i + \beta_4 A_i + \varepsilon_i.$$

Potential outcomes include:

- extraction cost;
- drilling failure;
- recovery rate;
- profitability;
- equipment lifetime.

Expected signs for extraction cost are

$$\beta_1 > 0, \quad \beta_2 > 0, \quad \beta_3 > 0.$$

For astronomical projects,

$$Y_j^A = \gamma + \delta_1 D_j + \delta_2 \tau_j + \delta_3 E_j + \delta_4 A_{AI,j} + u_j.$$

Outcomes may include mission cost, success probability, communication quality, or expected net present value.

43 Survival Analysis

Mission and equipment failure can be modeled using hazard functions.

Let

$$h_G(t|d, T, P)$$

denote geological equipment hazard.

A proportional-hazards specification is

$$h_G = h_{0G}(t) \exp(\beta_d d + \beta_T T + \beta_P P).$$

For space missions,

$$h_A = h_{0A}(t) \exp(\gamma_D D + \gamma_R R_{\text{rad}} + \gamma_\tau \tau).$$

These models provide statistical estimates of the effective frontier.

44 Bayesian Frontier Estimation

The frontier itself is uncertain.

Let

$$\theta_G = (T_{\max}, P_{\max}, A_G, \dots)$$

and

$$\theta_A = (v_{\max}, A_A, H_{\text{rad}}, \rho, \dots).$$

Given data $\mathcal{D}$,

$$p(\theta|\mathcal{D}) \propto p(\mathcal{D}|\theta)p(\theta).$$

The induced posterior distributions are

$$p(D_G^*|\mathcal{D})$$

and

$$p(D_A^*|\mathcal{D}).$$

Policy should therefore depend not only on expected frontiers but on frontier uncertainty.

45 Machine Learning Formulation

Suppose the state vector is

$$s_t = (X_G, K_G, K_A, A_G, A_A, D_G^*, D_A^*, E, P, T, \text{political state}).$$

The action vector is

$$a_t = (I_G, I_A, I_G^R, I_A^R, q_G, M, C).$$

The reward is

$$r_t = U(C_t) - \Psi(E_t) - C_{\text{risk},t}.$$

The value function satisfies a Bellman equation

$$V(s) = \max_a \{r(s, a) + \beta \mathbb{E}[V(s')|s, a]\}.$$

This representation permits reinforcement-learning approximations to the dynamic planner problem.

46 Multi-Agent Planetary Equilibrium

For N nations, let agent i choose

$$a_i = (I_{Gi}, I_{Ai}, M_i, R_i).$$

State transitions depend on all agents:

$$s_{t+1} = F(s_t, a_{1t}, \dots, a_{Nt}, \varepsilon_{t+1}).$$

Each government maximizes

$$V_i(s) = \max_{a_i} \mathbb{E} \left[\sum_{t=0}^{\infty} \beta_i^t r_i(s_t, a_{it}, a_{-it}) \right].$$

This becomes a stochastic dynamic game.

Cooperative solutions maximize

$$\sum_i \omega_i V_i,$$

whereas noncooperative solutions seek Markov-perfect or other strategic equilibria.

47 Computational Algorithm

A numerical solution can proceed by nested frontier estimation and equilibrium iteration.

Pseudo-code

INPUT:

- planetary radius R
- geological data
- astronomical data
- technology parameters
- preferences
- resource stocks
- national states

INITIALIZE:

- DG
- DA
- prices
- capital allocation
- labor allocation
- research allocation

REPEAT:

1. Estimate geological frontier:
 - evaluate temperature
 - evaluate pressure
 - evaluate material limits
 - evaluate extraction profitability
 - DG <- minimum feasible frontier
2. Estimate astronomical frontier:
 - evaluate propulsion feasibility
 - evaluate communication delay
 - evaluate biological survival
 - evaluate causal restrictions

```

        evaluate discounted project value
        DA <- minimum effective frontier

3. Solve production decisions.

4. Update geological scarcity rents.

5. Update delayed astronomical prices.

6. Allocate labor and capital.

7. Allocate research between inward
   and outward frontier expansion.

8. If multiple nations:
   solve strategic best responses.

9. Update AI technology.

10. Update environmental externalities.

11. Update welfare.

UNTIL:
    prices, allocations, frontiers,
    and value functions converge

OUTPUT:
    equilibrium allocation
    DG*
    DA*
    shadow prices
    welfare
    policy variables

```

48 Numerical Architecture

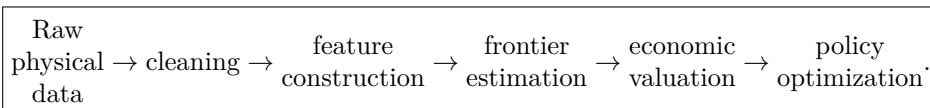
A practical computational implementation can consist of five modules:

1. **Geophysical module:** temperature, pressure, geology, mineral density, and drilling constraints.
2. **Astronomical module:** distances, trajectories, propulsion, communication, radiation, and causal constraints.
3. **Economic module:** production, preferences, prices, capital, resource scarcity, and discounting.
4. **Strategic module:** states, security competition, conflict, cooperation, and infrastructure control.
5. **AI module:** forecasting, frontier estimation, dynamic optimization, and autonomous control.

The modules communicate through a common state vector.

49 Data Science Pipeline

A reproducible empirical pipeline is:



Geological data may include:

- borehole depth;
- temperature;
- pressure;
- rock composition;
- extraction cost;
- equipment failures;
- commodity prices.

Astronomical data may include:

- mission distance;
- velocity;
- energy consumption;
- signal latency;
- radiation exposure;
- mission success;
- launch cost;
- scientific value.

50 A Causal Network Representation

Figure 2 summarizes the main economic mechanisms.

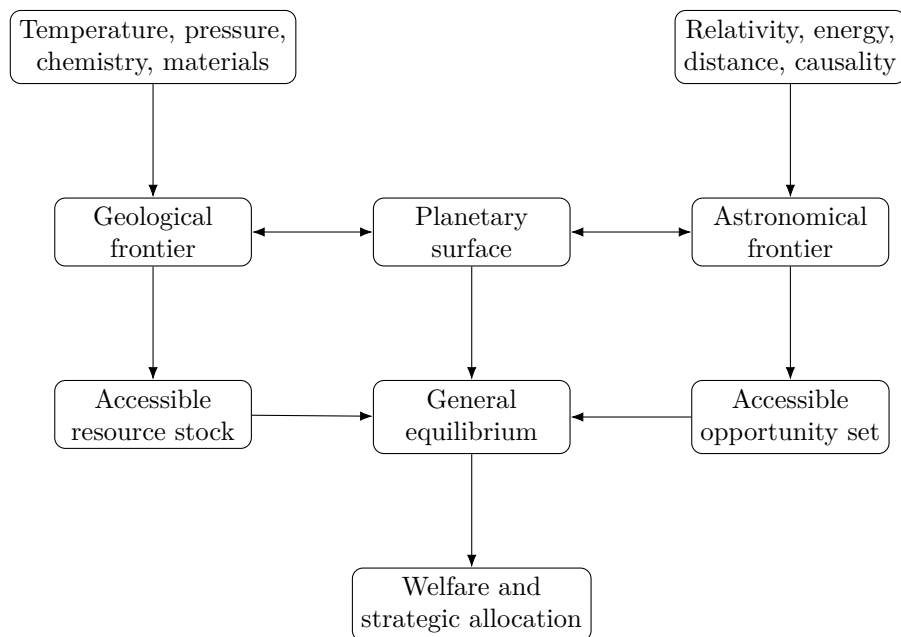


Figure 2: Causal architecture of Relativistic-Thermodynamic General Equilibrium.

51 Comparative Physical Constraints

Table 2: Physical and economic scarcity mechanisms

Domain	Constraint	Economic consequence
Geology	Temperature	Higher operating cost
Geology	Pressure	Equipment and structural limits
Geology	Chemistry	Corrosion and material degradation
Geology	Depth	Energy and logistics costs
Geology	Exhaustibility	Scarcity rent
Astronomy	$v < c$	Positive travel time
Astronomy	Relativistic energy	Rising propulsion cost
Astronomy	Signal speed	Information delay
Astronomy	Radiation	Survival and reliability risk
Astronomy	Distance	Discounting and governance cost
Both	Capital scarcity	Opportunity cost
Both	Knowledge scarcity	Research value
Both	Political risk	Strategic premium

52 Policy Instruments

A government may choose

$$\mathcal{P} = \{\tau_G, \tau_A, s_G, s_A, R_G, R_A, M, Z\},$$

where τ denotes taxes, s subsidies, R research expenditure, M security expenditure, and Z regulation. Efficient policy should account for:

1. exhaustible-resource scarcity;
2. environmental externalities;
3. scientific spillovers;
4. national-security externalities;
5. intergenerational effects;
6. international spillovers;
7. catastrophic risk.

The policy problem is therefore intrinsically second-best.

53 International Cooperation

Both domains can generate global public goods.

Geological research can improve:

- earthquake understanding;
- climate reconstruction;
- groundwater management;
- hazard prediction.

Astronomy can improve:

- planetary-defense knowledge;

- navigation;
- climate observation;
- fundamental science;
- communication infrastructure.

Let country i contribute R_i to international science.
Global knowledge is

$$A = \sum_{i=1}^N \eta_i R_i.$$

If each country captures only fraction $\alpha_i < 1$ of the global benefit, decentralized equilibrium generally underprovides research.

This creates a conventional rationale for international scientific cooperation.

54 Planetary Defense

Astronomical knowledge has an insurance interpretation.

Let catastrophic extraterrestrial loss be L with probability p .
Planetary-defense investment I_D reduces probability to

$$p = p(I_D), \quad p'(I_D) < 0.$$

Expected loss is

$$EL(I_D) = p(I_D)L.$$

Optimal investment satisfies

$$-p'(I_D)L = 1$$

when investment cost is normalized to one per unit.

Thus astronomy produces not only scientific output but catastrophic-risk reduction.

55 Geological Catastrophic Risk

Geological knowledge likewise produces insurance benefits through improved understanding of:

- earthquakes;
- volcanoes;
- landslides;
- groundwater;
- subsidence;
- geochemical hazards.

If geological research I_H reduces expected disaster loss,

$$EL_G = EL_G(I_H),$$

with

$$EL'_G(I_H) < 0,$$

then optimal research includes the avoided-loss term.

Thus both inward and outward science have insurance value.

56 The Two-Sided Insurance Principle

The planet faces hazards from both directions.

Interior hazards include

$$\mathcal{H}_G = \{\text{seismic, volcanic, geochemical, subsurface}\},$$

whereas exterior hazards include

$$\mathcal{H}_A = \{\text{asteroidal, solar, radiative, orbital}\}.$$

Scientific investment therefore insures civilization against hazards arriving from both sides of the planetary interface.

57 Information Theory

Finite communication capacity adds another constraint.

Let channel capacity be $C(D)$.

Suppose

$$C'(D) < 0$$

after accounting for signal attenuation, noise, power, bandwidth, and engineering limitations.
The useful information flow is bounded by

$$I(D) \leq C(D).$$

Consequently, distant economies may face not merely delayed information but limited information throughput.

This further encourages local autonomy and compressed sufficient-statistic communication.

58 Distributed Computation

A spatially extended civilization resembles a distributed computer system with nonzero message latency.

Each economic node i maintains local state

$$s_i(t).$$

Messages from j arrive after

$$\tau_{ij} > 0.$$

Hence consensus protocols must tolerate delayed and potentially stale information.

This analogy suggests importing concepts from distributed systems:

- asynchronous consensus;
- fault tolerance;
- local state estimation;
- eventual consistency;
- distributed optimization.

A relativistic economy should therefore be modeled more like an asynchronous network than a single instantaneous auctioneer.

59 A Distributed Price Algorithm

A stylized delayed price update is

$$p_i^{(k+1)} = p_i^{(k)} + \eta Z_i \left(p_i^{(k)}, \{p_j^{(k-\ell_{ij})}\}_{j \neq i} \right),$$

where Z_i is excess demand and ℓ_{ij} is communication delay measured in iterations.

Pseudo-code is:

```
for each market i in parallel:

    receive delayed prices from reachable markets

    estimate current remote states

    compute local excess demand

    update local price

    transmit updated price

    execute only causally feasible trades

repeat until delayed-network convergence
```

The convergence problem is therefore a delayed distributed fixed-point problem.

60 Thermodynamic Interpretation of Economic Cost

Economic production ultimately requires physical transformations.

Let useful work be W_u and energy input be E .

Efficiency is

$$\eta = \frac{W_u}{E}.$$

The second law of thermodynamics prevents arbitrary conversion efficiency in real physical processes.

Therefore frontier expansion necessarily consumes energy and produces entropy.

This observation places economic growth inside physical feasibility rather than treating production as an abstract transformation unconstrained by physics.

61 Energy as the Common Numeraire of Frontier Expansion

Both inward and outward expansion require energy.

Let

$$E_G(D_G)$$

denote cumulative energy required for geological access and

$$E_A(D_A)$$

energy required for astronomical access.

Generally,

$$E'_G(D_G) > 0,$$

and

$$E'_A(D_A) > 0.$$

The planner therefore faces

$$E_G(D_G) + E_A(D_A) + E_C \leq \bar{E}.$$

The energy shadow price λ_E links the two frontiers.

This creates a direct general-equilibrium channel between geology and astronomy even if their outputs are otherwise technologically independent.

62 Main Propositions

The framework yields the following core conclusions.

1. **Surface-interface proposition.** The spherical surface is a scarce interface between inward and outward opportunity domains.
2. **Geological accessibility proposition.** The economically relevant geological depth can be strictly smaller than the planetary radius because thermodynamic, mechanical, chemical, technological, and economic constraints bind before the center.
3. **Astronomical accessibility proposition.** Astronomical economic accessibility is constrained by propulsion, energy, biology, discounting, communication delay, causality, and cosmology even without assuming a finite spatial edge.
4. **Relativistic information proposition.** Spatially separated markets cannot exchange newly generated information instantaneously.
5. **Delayed-equilibrium proposition.** Sufficiently spatially extended equilibrium should be represented using delayed, local, spacetime-indexed information rather than instantaneous global information.
6. **Research-allocation proposition.** Efficient frontier research equalizes risk-adjusted marginal social returns from expanding inward and outward accessibility.
7. **Autonomy proposition.** Increasing communication delay raises the relative value of local autonomous decision-making.
8. **Strategic-frontier proposition.** Both geological and astronomical infrastructure acquire geopolitical and military shadow values in addition to civilian productive value.

63 Limitations

The framework is intentionally general and should not be interpreted as a complete geophysical or relativistic model.

Several qualifications are important.

First, real planets are not perfectly spherical.

Second, geothermal gradients are heterogeneous and nonlinear.

Third, geological accessibility depends on local geology rather than radius alone.

Fourth, astronomical distance in cosmology cannot always be treated as a single Euclidean variable.

Fifth, special-relativistic expressions are local approximations when cosmological expansion and curved spacetime matter.

Sixth, an observable horizon, causal horizon, event horizon, and technological reach frontier are distinct concepts.

Seventh, the existence of finite economic accessibility does not prove that physical space is finite.

Eighth, human preferences, institutions, and technologies evolve endogenously.

These limitations motivate more detailed versions of the theory rather than invalidating the basic constrained-accessibility architecture.

64 Conclusion

A civilization living on a spherical planet occupies a two-sided economic boundary.

At every point on the surface,

$$-\mathbf{n}$$

points toward geological opportunity, while

$$+\mathbf{n}$$

points toward astronomical opportunity.

The local geometry is symmetric.

The physical economics is not.

Geological activity encounters increasing temperature, pressure, chemical hostility, material constraints, extraction costs, and finite stocks. Consequently,

$$D_G^* < R$$

under the conditions developed in this paper.

Astronomical activity encounters finite signal speed, propulsion energy, radiation, biological limitations, discounting, institutional delay, and relativistic causal structure. Consequently, the economically actionable astronomical frontier at a given date can be finite,

$$D_A^*(t) < \infty,$$

without requiring physical space itself to possess a finite edge.

The economically relevant domain is therefore

$$-D_G^*(t) \leq z \leq D_A^*(t).$$

The planetary surface lies between two distinct scarcity regimes:

$$\text{Thermodynamics} \longleftarrow \text{Geology} \longleftarrow S^2(R) \longrightarrow \text{Astronomy} \longrightarrow \text{Relativity and cosmology}.$$

This physical architecture has a direct general-equilibrium consequence. Capital, labor, energy, land, knowledge, political authority, and scientific research must be allocated between inward and outward frontier expansion.

Moreover, once the economy becomes sufficiently spatially extended, finite signal speed prevents literal instantaneous coordination. General equilibrium must then become a theory of interacting local markets embedded in spacetime, connected by delayed information, transportation, financial claims, strategic institutions, and autonomous decision systems.

Relativistic-Thermodynamic General Equilibrium therefore replaces the fiction of an economically unconstrained spatial domain with a more physical principle:

$$\text{Economic opportunity} = \text{physical existence} \cap \text{causal accessibility} \cap \text{technological feasibility} \cap \text{institutional feasibility} \cap \text{positive economic value}.$$

The resulting object is not an infinite economic universe, nor merely a finite planetary economy, but an endogenous and evolving economic shell whose boundaries are determined jointly by physics, technology, knowledge, institutions, preferences, and time.

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Glossary

Accessible geological stock

The subset of physically existing geological resources that can be reached economically and technologically.

Astronomical frontier

The maximum outward distance $D_A^*(t)$ belonging to the economically actionable opportunity set at time t .

Causal accessibility

The ability of two events or regions to influence one another consistently with relativistic causal structure.

Communication delay

The positive time required for information to propagate between spatially separated economic agents.

Distributed equilibrium

An equilibrium represented by interacting local markets with delayed information rather than an instantaneous global auctioneer.

Economic frontier

A boundary beyond which an opportunity has nonpositive value or is otherwise economically infeasible.

Geological frontier

The maximum economically accessible inward depth D_G^* .

Geothermal gradient

The rate at which temperature changes with geological depth over a specified region.

Hotelling rent

The intertemporal scarcity value associated with an exhaustible resource.

Inward normal

The direction $-\mathbf{n}$ from the planetary surface toward the interior.

Outward normal

The direction $+\mathbf{n}$ from the planetary surface toward extraterrestrial space.

Planetary economic shell

The economically accessible radial region

$$-D_G^*(t) \leq z \leq D_A^*(t).$$

Radial access

Economic access obtained by moving inward or outward approximately along a surface normal.

Radial structural transformation

A long-run reallocation of capital, labor, research, and production between inward and outward economic domains.

Relativistic arbitrage

Arbitrage conditions modified to account for nonzero transportation and information propagation times.

Relativistic–Thermodynamic General Equilibrium

A general equilibrium framework in which economic accessibility is jointly constrained by planetary thermodynamics, relativistic causality, technology, institutions, and economic valuation.

Retarded information

Information received after a positive physical propagation delay.

Shadow price of depth

The marginal welfare gain from relaxing the geological accessibility constraint.

Shadow price of distance

The marginal welfare gain from relaxing the astronomical accessibility constraint.

Surface-interface value

The economic value of planetary surface area as an access platform connecting surface activity with inward and outward opportunity domains.

Thermodynamic frontier

A technological or economic boundary generated by temperature, energy, entropy, pressure, phase behavior, or related physical constraints.

The End